

BRITISH PATENT LAW

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TABLE III.—ACREAGE, YIELD, AND PRICES OF THE WORLD'S CHIEF CROPS, 1913-1920.

Season.	Acreage.	Per cent on 1913.	Crop.	Yield per acre.	Liverpool Prices (pence per lb.).		
					Lowest.	Highest.	Average.
			Bales 500 lbs. approximately	Bales.	Middling.		
American.	Acres.						
1913-1914	37,458,000	..	14,609,968	·39	6·20	7·96	7·26
1914-1915	37,406,000	100	15,067,247	·40	4·25	6·50	5·22
1915-1916	32,107,000	86	12,953,450	·40	5·34	8·74	7·51
1916-1917	36,052,000	96	12,976,000	·36	8·12	19·45	12·33
1917-1918	34,925,000	93	11,912,000	·34	16·90	24·97	21·68
1918-1919	37,207,000	99	11,640,000	·35	15·24	24·77	19·73
1919-1920	33,960,000	91	11,000,000	·33			
			Bales 400 lbs.	Lbs.	No. 1 Fine Oomra.		
Indian.	Acres.						
1913-1914	25,020,000	..	5,065,000	81	4·70	6·56	5·87
1914-1915	24,595,000	98	5,209,000	85	3·94	5·00	4·46
1915-1916	17,746,000	71	3,738,001	84	4·75	6·90	6·09
1916-1917	21,745,000	87	4,502,000	83	7·10	18·30	11·00
1917-1918	25,188,000	100	4,000,000	64	15·50	20·36	18·78
1918-1919	20,497,000	82	3,671,000	72	13·97	23·00	18·12
1919-1920	24,250,000	97	4,700,000	78			
			Kantars.	Lbs.	F. G. F. Brown.		
Egyptian.	Feddans.						
1913-1914	1,723,094	..	7,684,172	444	8·15	10·45	9·44
1914-1915	1,755,270	102	6,490,221	369	6·30	8·30	7·34
1915-1916	1,186,004	69	4,806,331	406	7·50	11·90	10·42
1916-1917	1,655,512	96	5,111,080	310	11·60	31·50	21·56
1917-1918	1,677,310	97	6,307,618	375	28·56*	35·50	30·97
1918-1919	1,361,000	79	4,820,660	354	26·59*	30·19	27·85
1919-1920	1,573,662	91	6,000,000	380			

The figures in *italics* are estimates.

* Sakel.

BRITISH PATENT LAW

By Sir GEORGE CROYDON MARKS, M.P.

There is more justification for an institution like the Textile Institute to have under consideration the subject of British patent law than for any other industrial assembly that could be gathered together in this country. If there is one industry more than another that is dependent on, and has been maintained, developed, and kept in being by inventions, it is that of the industry with which you are associated. Unquestionably, the textile industry is, as indeed all industries are at the base, founded on improvements and developments in mechanical and other devices which go towards the betterment of the products you are engaged in manufacturing.

There is great misconception in the popular mind as to what a patent is intended to do and what it is primarily for. A great many people associate the idea in connection with a patent as being a kind of reward to a man for having discovered or accidentally arrived at something that other people were not lucky enough to arrive at beforehand. As a matter of fact, a patent is something in the nature of a bargain between the State and the individual who has become possessed of an improvement or new process which someone else is unacquainted with, and which he could, if he desired, use to his own advantage and personal benefit with the risk that somebody else might find out the same thing he was doing and then enter into

competition with him. But, by bargaining with the State, he says: "I am willing to give up my prospective sole and entire monopoly in this invention for all time, or for as long as I can keep it secret, provided you, the State, will give me your legal protection for a limited period of years against any other person copying this, if I explain fully what my invention consists of." The State then says: "I will do this on one condition—that in regard to your invention or process you have discovered, you particularly describe it in detail in a public specification or document, so that at the end of the period when your monopoly ceases the industry itself shall be able by what you have so described in your specification to freely benefit by, and publicly thereafter use, what you have discovered."

The nature of the bargain between the State and the individual is, in effect, that the individual is willing to give up his personal monopoly for a possibly problematically uncertain period, in return for an entire monopoly for a limited period under protection from the State against piracy or copying by other people.

Mr Edison has been probably more associated with inventions than any other man in the world, and his definition of the results accruing to the public, as compared with those to the inventor, would be borne out by any inventor in this country. Mr. Edison

declares that the inventor gets two per cent of the benefit and the public 98 per cent from any useful invention. I do not hesitate to say that that is absolutely correct from my own long experience in association with important inventions and valuable commercial discoveries.

What I have just said is to clear away what might be misconception as to the idea of a monopoly. We are considered to be a free-trade country. Politics do not and cannot enter into the question of patents, for the granting of a patent is for something that the world knows nothing about apart from the inventor's discovery or production. You cannot have free trade in something that does not publicly exist, and, therefore, when a person comes along with an invention which is his own device, produced after years of thought and great expense, with many failures and few successes—when that man comes along, there is no such thing as having freedom in respect to its manufacture, for others know nothing of it, and, presumably, would not know if the inventor keeps it a secret. Therefore, when the State says "I will give you a monopoly," there is no question of violation of free trade, for it is a personal production in which others have had no part in producing.

The granting of all patents is something apart altogether from the theory of free trade, as such freedom is only possible in commodities which everyone can get hold of, or is able to freely make or sell. The granting of the patent has always for its primary object the benefit of the State and not the benefit of the individual alone. In the forefront of every patent ever granted, since the time of King James in 1624, there is set forth a condition associated with and declared on the document itself that the Royal Letters Patent, or open letter from the Crown, is only granted because it is believed to be for the public good. Therefore, the granting of the patent is publicly proclaimed to be for the public good and not simply a reward or benevolent gift to the inventor. When that is in our minds, we see that the great feature of the policy of granting patents for inventions is the development of national resources, and not a kind of prize for the clever person. In the very forefront of every patent, the King recites that which has prompted him to make the grant, in the following terms:—

GEORGE V., BY THE GRACE OF GOD, of the United Kingdom of Great Britain and Ireland and of the British Dominions beyond the Seas King, Defender of the Faith, Emperor of India : To all to whom these presents shall come greeting : WHEREAS H. T. of hath declared that he is in possession of an invention for " " that he is the true and first inventor thereof, that the same is not in use by any other person to the best of his knowledge and belief ; AND WHEREAS the said inventor hath humbly prayed that a patent might be granted unto him for the sole use and advantage of his said invention: AND WHEREAS the said inventor hath by and in his complete specification, particularly described the nature of his invention ; AND WHEREAS we being willing to encourage all inventions which may be for the public good are graciously pleased to condescend to his request.

Provided that these our Letters Patent are on this condition, that if at any time during the said term it be made to appear to us that this our grant is contrary to law or prejudicial or inconvenient to our subjects in general, or that the said invention is not a new invention as to the public use and exercise thereof within our United Kingdom, or that the said patentee is not the first and true inventor thereof within this realm, these our Letters Patent shall forthwith determine and be void to all intents and purposes, notwithstanding anything hereinbefore contained.

There are two systems in connection with the granting of patents. One system, adopted by some countries, is simple registration : that is, a man deposits a specification describing that which he has done, illustrating and showing what he considers he is entitled to claim as his new invention, and this description is registered without any official investigation or examination as to the claims made. He has to pay annual taxes to the State because of the monopoly thus granted. The States that have thus registered, grant no particular benefit beyond assuring him that they will protect the invention in their Courts of Law should he in such Courts establish his title and prove that his invention was new or novel ; but he gets no guarantee that the same invention has not been previously patented or registered a dozen times over. That is the system in France, Spain, and other countries to-day. You can obtain a Patent for the same thing that someone else obtained a patent for last week. That used to be the system in England. Now, however, in England, under the existing law, you apply for a patent and it is after an official examination that you get it ; examination is also made, more fully, in America and Germany. In England, the examination is limited to the British specifications of patents that have been published and granted during the preceding fifty years, so that a man cannot now say to the Government—"You are a fraudulent institution in that you take my money and fees for something in respect of which you have already been paid other fees from previous persons years ago."

The present system of the British Patent Office is that you deposit the specification, and in effect pay for examination, the examiners' duty being to search and then say whether it has been patented in England before. If they think, as a result of their search, that it has, they give particulars of the specifications discovered and the patentee must amend the specification, so that it becomes clear or free of the earlier patents. That is good in its way, but it sometimes makes a man think, when he gets the patent, that he can then go round and vigorously threaten people to leave off doing that which they are engaged upon, or stop what he alleges is an infringement of his patent. Often, however, he is met with the reply that the American people, or others abroad, have previously patented it and the foreign specification was previously known to them and published in this country. It is not, in legal phrase, a "new" invention if it was known in any public book or document existing in this country before. In England, however, only the British patents are searched or examined by the Patent Office Examiner, and the result is that, every

day, patents are granted in England that would fail in the first half hour they went into a Court of Law, because they have been anticipated by documents previously published in this country.

The American system of examination is wider, for it provides that, on an application being made for a patent, not only the American specifications are to be examined, but every other specification that they happen to be familiar with, and a man may then be told that it is something previously patented, not in America but elsewhere, and that he is therefore seeking to obtain a patent for that which is not novel or new. Then, he has to amend his specification to get clear of the citations, and, presumably, he gets something which he thinks is quite valid.

On the face of it, this looks excellent. It looks as though the American gets a solid, guaranteed bit of property. But it is not so guaranteed; neither is it safe to consider it valid. It really only means that the Examiners have not found or disclosed anything exactly like it. An expert, or the man himself, may find many things, on a rigorous search, which the Examiners have not. Consequently, this assumed security provokes more litigation over patents in America in one month than there is in this country in twelve months. The great difficulty with the American patentee—and there is an enquiry going on now in regard to the matter—is that the man is over-confident when he gets an American patent, and he finds, frequently, when he gets into Court, that the Court does not consider it any more valid than does an English Court an English patent. Judged by results of the two systems, over-confidence arises in America; in England, there is no over-confidence, but there is some kind of half assurance which makes a man probably more careful so far as instituting litigation is concerned.

In Germany, the same kind of general or full examination is pursued before grant. The German Government, however, is far more particular in examination than this country or America. If there is any possible chance of the German Examiner killing or whittling down an application for a patent by citing old publications of any kind, however far-fetched, he will kill it, the idea apparently being that if he does so, it will then be free for the benefit of the home German industry. The search is most rigorous and the German patent is therefore frequently considered as being a kind of hall-mark of originality that a man can use in his own country, or elsewhere, by reason of the rigid examination.

Having been granted his British Patent at a total cost in Government fees of Five Pounds, at the end of four years the patentee has to pay a renewal fee of £5 to cover the fifth year, and thereafter at the end of each year an annual renewal fee is payable increasing in amount by one pound per annum up to the fourteenth or last year, when fourteen pounds is payable to the British Patent Office. The Patentee, at the end of four years, having to pay £5 in order to keep his patent alive for the next year, asks himself if it is worth it, and it is at this stage that more than 50 per cent of

the patents fall out as not being worth a further five pounds to the patentee.

In America, a Patentee gets his patent at a Government fee cost of \$35, and that lasts for 17 years. He has no other fees to pay. It is undoubtedly the conviction of all American Attorneys that if he had to pay only a further \$10 at the end of four years he would not pay it to keep the patent alive another year. The result of this long term grant is that there are thousands of purely impracticable paper anticipations and specifications that are veritable dogs-in-the-manger to hold back other good inventions, because those skeleton ideas have never been worked but are still kept alive even though their authors neither value nor want them. When, however, someone else comes along to make an improvement in the same industry, he cannot work it commercially because there is an old piece of paper with some semblance of a notion that might conflict with his own details, patented before his. If it is pointed out that as the invention has not been worked it cannot be of value, the later inventor, when negotiating with the earlier owner, will find that this forgotten idea is the most valuable thing on which he had all his hopes for future happiness and prosperity.

I do not hesitate to say it is a common inducement and occasion for blackmail. Those patents with no life cannot be cleared out of the way, and yet they are no real good. Of the 30,000 patents annually applied for in England, only some 48 out of every 1,000 are considered by their owners as being worth the annuities to keep them in being for the full term of 14 years. The remaining 952 that would not be dead if in America, are felt to be a nuisance in America and are commonly in the way of good developments. The experienced American people familiar with patents would like to have some kind of official system, or annual taxation, which would get rid of these paper things and remove from industry those inventions which, while they live without any benefit to their owners, block all possible discovery.

A patent, as already shown, is avowedly granted for the public good. If a man in this country does not use, nor intend to use, his invention, at the end of four years he can be required to work and use it in this country, or to allow a license to be secured by another who will so use it. The old system—which was attacked by the Act passed by Mr. Lloyd George when President of the Board of Trade in 1907—was such that the foreigner frequently obtained a patent in this country, not for the purpose of using it, but for the purpose of preventing anyone else making any use of it, whilst he made all the material or constructed the machine abroad, charging a higher price here for the imported goods because of the absurd patent protection he had obtained that prevented others from making or using it without his consent. This was thought to be remedied by the 1907 Act, but under the Act all foreigners can apply to the British Patent Office and ask for British patents to be revoked because they are not being worked or used here, and when the patent is so revoked, the foreigner, who is making the goods in his own country where no patent exists, can then

import them here without having to pay royalty to, or be hindered by, the claims of the British inventor whose patent has been revoked.

In the present Parliament, a new Patents Bill has passed the third reading in the House of Commons, and should it pass the House of Lords, will probably be law in three months' time.

It is interesting to note what has been published in Germany concerning the British Cotton Industry:—

"Although the origin and application of inventive genius is largely independent of known laws, and may provisionally be relegated to the domain of 'accident,' there are certain reasons which favour the cotton industry in the industrial race. Its concentration in South Lancashire and Staffordshire, as compared with the wide diffusion of the woollen industries, facilitated the rapid acceptance of new *methods and discoveries*. Moreover, the cotton industry being of later origin, and settling itself in unimportant villages and towns, had escaped the influence of official regulations and customs which prevailed in the woollen centres and proved serious obstacles to the introduction of new industrial methods."

As showing the dependence of the cotton industry on mechanical inventions, evidence was given before a House of Lords Committee, as far back as 1857, that "the present spinning machinery which we now use is supposed to be a compound of about eight hundred inventions. The present carding machinery is a compound of about sixty patents."

There is nothing fortuitous or mysterious in inventive energy. Necessity is its mother, which simply means that it moves along the line of least resistance. Men like Kay, Hargreaves, Arkwright, Cartwright, set their intelligence and industry to meet the several difficulties as they arose. Nearly all the great textile inventors were practical men, most of them operatives immersed in the details of their craft, brought face to face continually with some definite difficulty to be overcome, some particular economy desirable to make. Brooding upon these concrete facts, trying first one thing, then another, learning from the attempts and failures made by other practical men, and improving upon these attempts, they have at length hit upon some contrivance that will get over the definite difficulty and secure the particular economy. If we take any definite invention and closely investigate it, we shall find in nearly every case it has thus grown by small increments towards feasibility. Scientific men, strictly so called, have had very little to do with these great discoveries. Among the great textile inventors, Cartwright alone was a man leading a life of thought. When the spinning machinery was crippled in its efficiency by the crude methods of carding, Lees and Arkwright set themselves to apply improvements suggested by common sense and experience; when Cartwright's power-loom had been successfully applied to wool, Horrocks and his friends thought out precisely those improvements which would render it remunerative in the cotton trade.

The following is a Schedule showing the relation of the growth of the cotton industry as measured by

the consumption of raw cotton and against leading mechanical inventions:—

COTTON IMPORTED.		INVENTIONS, ETC.	
	lbs.		
1730	1,545,472	1730	Wyatt's roller-spinning (patented 1738).
		1738	Kay's fly-shuttle.
1741	1,645,031	1748	Paul's carding-machine (useless until improved by Lees, Arkwright, Wood, 1772-74).
		1764	Hargreaves' spinning-jenny (patented 1770), for weft only.
1764	3,870,392	1764	Calico-printing introduced into Lancashire.
		1768	Arkwright perfects Wyatt's spinning-frame (patented 1769), liberating cotton from dependence on linen warp.
1771 to 1775	4,764,589	1771	Arkwright's mill built at Cromford.
		1775	Arkwright takes patents for carding, drawing, roving, spinning.
		1779	Crompton's mule completed (combining jenny and water-frame, producing finer and more even yarn).
1781	5,198,775		
1785	18,400,384	1785	Cartwright's power-loom. Watt and Boulton's first engine for cotton mills.
1792	34,907,497	1792	Whitney's saw-gin.
1813	51,000,000	1813	Horrocks' dressing-machine.
1830	261,200,000	1830	The "Throstle" (almost exclusively used in England for spinning warp).
1832	287,800,000	1832	Roberts' self-acting mule perfected.
1841	489,900,000	1841	Bullough's improved power-loom.
			Ring spinning (largely used in U.S.A., recently introduced into Lancashire).

Between the years 1855-1906 the number of inventions in connection with spinning for which patents have been granted is 10,696.

The textile industry was founded and developed on, and is still maintained on, inventions. That being so, we are more concerned, so far as this national industry is concerned, in having a good patents system than any other industry. We have grown up on invention and progressed, and, having progressed, we want to see development. All developments of all industry have reference to four factors—invention, capital, labour, plant or establishments. Unless we have someone with enterprise, invention is useless; labour, thus, can only come in after invention has been made and capital invested. There is no particular reason why these four factors should not be intermeshed with a recognition of their interdependence one upon the other.

I do not hesitate to say that the new Patents Bill now before Parliament is a new charter in respect of all patents. It starts away by lengthening the life of a

patent from 14 to 16 years without extra fee, and those now granted for 14 years are to have an extra two years of life. Instead of having to pay a fee at the end of four years, the payment will arise at the end of six years, which will give a man a pretty good run to test or start his proposal. It is generally the first three or four years in the life of a patent which takes the heart out of a man, and if he was able to go on a few years longer, he might be successful. There is to be no renewal fee until the end of six years. That will make the new British system for a period of six years the cheapest patents system in the world. The examination will not be interfered with. As patents are granted for the public good, and the monopoly granted for the benefit of the national industry, this Bill provides that if any man abuses his monopoly, then after four years it is open to anyone to go to the Patents Office and say: "There has been an 'abuse of monopoly,' he is not working it and won't let anyone else, and I now want a license that I may work it." He may further say that instead of working it he is importing the articles. If the abuse is proved, the Comptroller can say that a license shall be granted upon fair royalty terms, which the Patent Office will then settle. It is thus provided that a patent shall be worked on a commercial scale without undue delay.

This is a great move forward in connection with British industry, and is going to have a very far-reaching effect. It means another thing—that when a man has been proved to have abused his monopoly rights, the Comptroller will endorse the patent "License as of Right," and the moment it is so endorsed, any person who sees it can apply for a license. He will get a license on royalty. If a man has his patent thus endorsed, that does not kill it; he will get reasonable royalty determined by the Patent Office after proper enquiry at which he will be heard.

Now take the other side. A man may be too poor to work his invention. The greatest number of inventions in connection with the textile industry have been produced by practical men, not by great thinkers and investigators, but men in the industry familiar with the requirements. Unfortunately, these men are not always owners; they have not got it in their power to try the invention, and are at the mercy of someone who will offer, say, to give £10 for it. If a man knows he cannot work his invention he can go to the Patent Office and ask for the patent to be endorsed "License of Right" at the beginning or at any time. The patent can be endorsed, and then anyone can apply for the license. The Patent Office would decide the royalty. The moment a man elects to have his patent endorsed, the Crown says: "I will only ask half the annual renewal fees hereafter from you." Thus, at the end of 6 years, under the new Bill when passed, instead of £5, £2 10s. only will be payable; at the end of 7 years £3 10s., and then at the end of 15 years £7 only instead of £14. This is a great move forward for the benefit of inventors who do not know what to do with their inventions, and it is a check upon some employers or others depriving men of their property. It is also an assurance that the foreigner shall no longer be able to have a

patent and cripple our industry by not working it here himself. The patent will be endorsed and will be kept alive. If the House of Lords is not too troublesome, this Bill ought to be law before the end of this year, and every existing patent will then be given two years additional life.

Further, supposing a man has a patent, and a man comes for a license—asks in fact, for an exclusive license which will make him feel safe in putting down his money for new plant—the Court has a right under the Bill to put him to the test, and say: "You shall have an exclusive license provided you undertake to spend the money you say or we think necessary." If he does not spend the money in the time named, the license can be revoked. That prevents selfishness and unfair exclusive dealing.

The Bill is the outcome of deliberations of the Board of Trade with different interests represented, and I really think it, as far as one can see, a very fine chance for the development we all so badly need in the matter of industrial reconstruction, without taking advantage of anyone who ought not to be taken advantage of. This is going to be of great benefit to industry.

I have said that, in America, patent litigation is a frightful expense. The British system of patent litigation has been criticised as wrong in principle and unjust to the inventor. Every specification has in it claims the man considers he is entitled to; he may have 10 claims. One little feature may be old, and because of that, if taken into Court at the present time, the patent fails. Bad in the little, it is bad in the whole. Patents frequently fail, not because a man's chief claims are found to be bad, but because some part is found to be old and that makes the whole patent fabric fall down.

Under this new Bill, a man will only have to sustain in the Court the claim he alleges is being infringed, and that will be of far-reaching effect. It is following the American system, which requires only the claim relied upon to be valid.

The charter of the new Patent Bill is longer life, freer working, less fees, no confiscation, no undue advantage of the ignorance of a man in reference to details of his specification, and I believe, myself, a new era is possible to inventors by giving them an outlet for their inventions by this system of licenses as of right.

DISCUSSION.

The PRESIDENT said he was particularly interested in the subject because he happened to be an inventor and a patentee. He greatly appreciated all that had been said, and he thought the new Bill would be a wonderful encouragement to invention and great justice to the inventor.

Mr. HAROLD POTTS, M.Sc. (Liverpool) proposed a vote of thanks to Sir Geo. Croydon Marks, and said the Institute was to be congratulated upon securing the services of a speaker of such distinction in his profession. The utterance was of the greatest importance. However, he was inclined to venture the view that the American system of examination was not quite so bad as Sir George had made out. Over-confidence was suggested, but he (Mr. Potts) thought it might

just as well be claimed that examination of ships by Lloyd's led to over-confidence on the part of the navigator. The reason why there was so much litigation in the States was not false confidence but a deeper difference in the national psychology. Americans took more kindly to the patent system as a whole. All classes in America appreciated the value and the improvements of patents. The provisions of our new Bill would do little good unless the public at large and the manufacturers in particular took more interest in the whole subject. In America, a board of directors could usually discuss intelligently questions of patent policy, and he was afraid it was not so in this country. We required freer discussion and more education, and that was why he appreciated the efforts which the

Research and Inventions Committee of the Textile Institute had made in regard to this subject. If every other Institute gave special consideration to the provisions of the new Bill, great good would be accomplished.

Mr. FRANK WRIGHT (Bolton) seconded, and advocated the regular examination of the patents abridgements for the section of industry engaged in.

Mr. LUKE THORNER (Burnley): Will the new Act apply to old patents?

Sir G. CROYDON MARKS: Yes.

The motion was heartily accorded, and Sir G. CROYDON MARKS returned thanks, and said he would be pleased to answer written questions sent through the Institute.

WHY OUR TRANSPORT RATES ARE THE HIGHEST IN THE WORLD

By A. W. GATTIE.

The value of everything depends upon its being where it is wanted, when it is wanted. It is the function of transport to bestow this value. Efficient transport is, therefore, a matter of supreme importance to the trade of the country. Our transport is slow, uncertain, and expensive. We pay twice, and more than twice, the sum which is paid for the service better rendered to our competitors. To continue such a policy, is to continue a policy of protection in favor of the foreigner.

The rate per ton per mile, for the United Kingdom—according to statistics published by Mr. Slason Thompson, Secretary of the Railway Managers' Association of Chicago—is 1.123d.; France 0.726d.; Germany 0.637d.; Holland 0.590d. The three last-named countries all have canals, whereas our own canal system has been largely strangled. The 1.123d. for the United Kingdom ought to be nearer 1.5d. but for the erroneous character of our Board of Trade returns in this connection. One of the chief sufferers owing to this excessive cost of transport is our agriculturist. Given cheap transport, we could grow our own food, whereas, owing to this abnormality in our transport conditions, we were only producing, at the time of outbreak of the war, 25 per cent of the food necessary to our existence.

It is manifest that the efficiency of transport must depend upon the efficiency of the instruments of transport, and these, as far as railways are concerned, are the railway goods waggon and the locomotive. Let us look at this matter of the life of a goods waggon. A waggon is only engaged in carrying goods on the track for six minutes out of 24 hours, and it is in detention, for one reason or another, for 23 hours and 54 minutes. The diagram which I am using to illustrate these ascertainment, was originally exhibited nearly ten years ago at the London Chamber of Commerce, and it was recently sworn to by me before the Committee appointed by Government to enquire into my system. Learned counsel employed by the

railway companies thought it wise, apparently, not to dispute its accuracy.

There are 1,500,000 of these waggons in the United Kingdom and, at pre-war prices, they cost £100 each to build, and £5 a year each to maintain. They idly occupy about £300,000,000 worth of land, where they are senselessly pushed about by a locomotive for about 97% of their time. This senseless pushing about is called "shunting." It injures goods, smashes rolling stock, and absorbs 80% of the energies of the locomotive.

I now turn to the matter of the life of a locomotive. It is only engaged in the actual haulage of trains for two hours a day, and most of its energies are expended in pushing waggons backwards and forwards and smashing them. I have been told that this expensive folly must continue, because large and powerful interests are concerned in the repairing of these railway waggons, but I am quite unmoved by this argument. We don't tell our servants to fall down the kitchen stairs with our best dinner service because it will benefit the crockery shop. The average locomotive is half its time in the shed. In a week of 168 hours, it is engaged in hauling—the real work it is intended for—for 14 hours, whilst in shunting and marshalling it is occupied for 62 hours or 36.9% of its time. Moreover, there are a lot of privately-owned locomotives which are doing this shunting work almost exclusively. We want 300% increase in the efficiency of the locomotive.

The excuse for pushing waggons backwards and forwards, see-saw fashion, is, that it is the way to sort them. That won't do. What is really required is to sort the goods. To use a railway goods waggon, which can only move in one dimension, i.e., backwards and forwards, is manifestly absurd. A goods waggon is not a sorting machine. When any ordinary man wants to do a job, the first thing he looks for is the proper tool, instrument, or machine. You textile people know something about machinery. You know what its introduction and improvement have meant to your industry. Well, here we have, in 1919, a great